4CSPL1012: PROBLEM SOLVING USING PYTHON

Module 2: Basics of Python – Part II

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Credits: Respective owners

Codes: Click on this link to get Google colab file

Type Conversion (Implicit & Explicit)

Type conversion = changing one data type into another

Implicit Type Conversion (Automatic)

- Done automatically by Python

Example:

```python
a = 5       # int
b = 2.5     # float

c = a + b   # int → float (automatic)
print(c)    # 7.5
```

Explicit Type Conversion (Manual)

👉 Done by programmer using functions

Example:

```python
x = "10"

y = int(x)    # string → int
print(y + 5) # 15
```

Summary:

Type	Done By	Example
Implicit	Python	int → float
Explicit	Programmer	str → int

Explicit Type Conversion

Conversion Type	Function Used	Example	Output
String → Integer	`int()`	`int("10")`	10
String → Float	`float()`	`float("10.5")`	10.5
Integer → Float	`float()`	`float(5)`	5.0
Float → Integer	`int()`	`int(10.8)`	10
Integer → String	`str()`	`str(100)`	"100"
Float → String	`str()`	`str(5.5)`	"5.5"
Boolean → Integer	`int()`	`int(True)`	1
Integer → Boolean	`bool()`	`bool(0)`	False

Note: “Explicit conversion gives full control to the programmer to change data types.”

Operators and Expressions

- 👉 Operators = symbols used to perform operations
- 👉 Expression = combination of variables + operators

Types of Operators:

Arithmetic Operators:

Operator	Meaning	Example
+	Addition	5 + 2
-	Subtraction	5 - 2
*	Multiplication	5 * 2
/	Division	5 / 2

Relational Operators

Operator	Meaning
>	Greater than
<	Less than
==	Equal to

Logical Operators

Operator	Meaning
and	Both true
or	Any true
not	Opposite

And many more – will be learnt through examples

Examples: Arithmetic Operators

```python
a = 10
b = 3

print("Addition:", a + b)          # 13
print("Subtraction:", a - b)       # 7
print("Multiplication:", a * b)    # 30
print("Division:", a / b)          # 3.33
print("Modulus:", a % b)           # 1
print("Power:", a ** b)            # 1000
print("Floor Division:", a // b) # 3
```

Example: Relational Operators

```python
x = 5
y = 10

print(x > y)    # False
print(x < y)    # True
print(x == y)   # False
print(x != y)   # True
print(x >= 5)   # True
print(y <= 10) # True
```

Example: Logical Operators

```python
a = 5
b = 10

print(a > 0 and b > 5)    # True
print(a > 0 or b < 5)     # True
print(not(a > 0))         # False
```

4. Assignment Operators Assign values

```
x = 5

x += 3    # x = x + 3
print(x) # 8

x *= 2    # x = x * 2
print(x) # 16
```

Expressions (Combination of Operators + Values)

Expression = combination that produces a result

Operator Precedence Table

Precedence Level	Operators	Description
Highest	`()`	Parentheses
2	`**`	Exponent (power)
3	`*, /, //, %`	Multiplication, Division
4	`+, -`	Addition, Subtraction
5	`<, >, <=, >=, ==, !=`	Comparison
6	`not`	Logical NOT
7	`and`	Logical AND
Lowest	`or`	Logical OR

Example:

```python
age = 25

print(age > 18 and age < 60)
```

Output: `True`

Explanation:

Step	Condition	Evaluation	Result
1	`age = 25`	Assign value	—
2	`age > 18`	`25 > 18`	True
3	`age < 60`	`25 < 60`	True
4	`True and True`	Apply `and` operator	True

Example:

```
print(2 ** 3 * 2)
```

Output: 16

Operator Precedence: ** first → then *

Try for yourself:

1.
```
print(10 + 5 * 2)
```

2.
```
print((10 + 5) * 2)
```

3.
```
print(20 / 5 + 3)
```

4.
```
print(5 > 3 and 2 < 1)
```

5.
```
print(not(5 > 3))
```

Decision Constructs:

Decision constructs allow a program to **choose different paths based on conditions**

"Decision making = YES / NO logic"

Types:

1. `if`
2. `if-else`
3. `if-elif-else`

Before we discuss these, it is much needed to get understanding of **Indentation** in Python.

Indentation in Python

👉 **Indentation = spaces at the beginning of a line**

👉 It defines **block of code (group of statements)**

"In Python, indentation replaces curly braces {} used in other languages."

Why Indentation is Important?

- Defines **which statements belong to which block**
- Mandatory in Python (not optional)
- Improves readability

```
num = 10

if num > 0:
print("Positive number")    # ❌ Error
```

👉 ❗ Output:
IndentationError

1. if Statement

👉 Executes block only if condition is **True**

Program 1: Check Positive Number

```python
num = int(input("Enter number: "))

if num > 0:
    print("Positive number")
```

Output:

```
Enter number: 10
Positive number
```

Program 2: Check Divisible by 5

```python
num = int(input("Enter number: "))

if num % 5 == 0:
    print("Divisible by 5")
```

Output:

```
Enter number: 45
Divisible by 5
```

Program 3: Check Voting Eligibility

```python
age = int(input("Enter age: "))

if age >= 18:
    print("Eligible to vote")
```

Output:

```
Enter age: 19
Eligible to vote

Process finished with
```

Program 4: Check Pass Condition

```python
marks = int(input("Enter marks: "))

if marks >= 40:
    print("Pass")
```

Output:

```
Enter marks: 55
Pass
```

Common Errors in Selection Statements

Most common errors in selection statements are caused by incorrect indentation

```
radius = -20

if radius >= 0:
    area = radius * radius * math.pi
print("The area is", area)
```

(a) Wrong

```
radius = -20

if radius >= 0:
    area = radius * radius * math.pi
    print("The area is", area)
```

(b) Correct

2. if-else Statement (Two Conditions)

- One block executes if True, another if False, syntax is as shown in below example

Program 1: Even or Odd

```python
num = int(input("Enter number: "))

if num % 2 == 0:
    print("Even")
else:
    print("Odd")
```

Output:

```
Enter number: 23
Odd
```

Program 2: Positive or Negative

```python
num = int(input("Enter number: "))

if num >= 0:
    print("Positive")
else:
    print("Negative")
```

Output:

```
Enter number: -12
Negative
```

Program 3: Largest of Two Numbers

```python
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

if a > b:
    print("a is largest")
else:
    print("b is largest")
```

Output:

```
Enter first number: 13
Enter second number: 22
b is largest
```

Program 4: Check Leap Year

```python
year = int(input("Enter year: "))

if year % 4 == 0:
    print("Leap Year")
else:
    print("Not a Leap Year")
```

Output:

```
Enter year: 2026
Not a Leap Year
```

3. if-elif-else Statement (Multiple Conditions)

👉 Used when there are **multiple choices**

Program 1: Grade Calculation

```python
marks = int(input("Enter marks: "))

if marks >= 90:
    print("Grade A")
elif marks >= 60:
    print("Grade B")
elif marks >= 40:
    print("Grade C")
else:
    print("Fail")
```

Output:

```
Enter marks: 73
Grade B
```

Program 2: Largest of Three Numbers

```python
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))

if a > b and a > c:
    print("a is largest")
elif b > c:
    print("b is largest")
else:
    print("c is largest")
```

Output:

```
Enter first number: 11
Enter second number: 4
Enter third number: 9
a is largest
```

Program 3: Simple Calculator

```python
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
op = input("Enter operator (+, -, *, /): ")

if op == "+":
    print(a + b)
elif op == "-":
    print(a - b)
elif op == "*":
    print(a * b)
elif op == "/":
    print(a / b)
else:
    print("Invalid operator")
```

Output:

```
Enter first number: 14
Enter second number: 3
Enter operator (+, -, *, /): *
42
```

Program 4: Electricity Bill Calculation

```python
units = int(input("Enter units: "))

if units <= 100:
    bill = units * 1
elif units <= 200:
    bill = units * 2
else:
    bill = units * 3

print("Bill amount:", bill)
```

Output:

```
Enter units: 225
Bill amount: 675
```

Summary:

Type	Use Case	Example
if	Single condition	Check positive
if-else	Two conditions	Even/Odd
if-elif-else	Multiple conditions	Grades

Practice Problems:

Link to download the practice problems